Juneseo Chang

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http://jschang0215.github.io | Google Scholar

Research Interest

Computer Architecture, **Memory Systems**, **Emerging Memory Technologies**, and **HW/SW Co-design**

Education

Stanford University Stanford, California
Ph.D. in Electrical Engineering 2026.09 - Present

- Advisor: Prof. Thierry Tambe

Seoul National University (SNU) Seoul, Republic of Korea
B.S. in Computer Science and Engineering 2021.03 - 2026.08

- **GPA: 4.16/4.30** (**Rank: 3/110**), **Summa Cum Laude**.
- Including 18 months of mandatory military service at Republic of Korea Army (2024.07 - 2026.01).

Publications

1. **Juneseo Chang**, Wanju Doh, Yaebin Moon, Eojin Lee, and Jung Ho Ahn, "IDT: Intelligent Data Placement for Multi-tiered Main Memory with Reinforcement Learning", *International Symposium on High-Performance Parallel and Distributed Computing (**HPDC**)*, 2024
2. **Juneseo Chang** and Daejin Park, "Searching Optimal Compiler Optimization Passes Sequence for Reducing Runtime Memory Profile using Ensemble Reinforcement Learning", *International Conference on Embedded Software (EMSOFT) Work-in-Progress Track*, 2023
3. **Juneseo Chang**, Sejong Oh, and Daejin Park, "Accuracy-Area Efficient Online Fault Detection for Robust Neural Network Software-Embedded Microcontrollers", *International Conference on Embedded Software (EMSOFT) Work-in-Progress Track*, 2022
4. **Juneseo Chang**, Myeongjin Kang, and Daejin Park, "Low-Power On-Chip Implementation of Enhanced SVM Algorithm for Sensors Fusion-Based Activity Classification in Lightweighted Edge Devices", *Electronics*, 2022

Research Experience

Tambe Lab, Stanford University 2026.03 - Present

- Research on a retention-aware heterogeneous SRAM/gain-cell eDRAM LLC for serverless platforms (under review).
- Research on profile-guided composition and placement for heterogeneous SRAM/gain-cell eDRAM accelerator scratchpads (under review).
- Research on retention-aware number formats and a reconfigurable multi-precision systolic array in HLS for video diffusion transformer accelerators.

Scalable Computer Architecture Lab, SNU 2023.01 - 2024.06

- Advisor: Jung Ho Ahn
- Research on RL-based OS-level multi-tiered memory management, achieving 2.08× speedup over Linux kernel (**HPDC 2024, Acceptance rate: 17%**).

Relevant Coursework

Hardware System Design, Scalable High Performance Computing, Mobile Computing and Applications, Mathematical Foundations of Deep Neural Networks, Advanced Computer Architecture (Graduate), Advanced Compilers (Graduate), Data Communications, Abstract Algebra

Skills

Programming: `C++`, `C`, `Python`, `Verilog`, `Java`, `Bash`, `CUDA`, `PyTorch`, `TensorFlow`, `TFLite`, `RLLib`

Tools: `SST`, `QEMU`, `SCALE-Sim`, `Catapult HLS`, `NVIDIA Nsight`, `Amaranth`, `LLVM`, `Linux` kernel, `DAMON`

Languages: English (TOEFL MyBest Score 111/120), Korean

Scholarships and Awards

Semiconductor Specialized University Scholarship, SNU 2024.03 - 2026.08
$8,000 over three semesters.

Presidential Science Scholarship, Korea Student Aid Foundation 2021.03 - 2026.08
Full tuition and stipend for eight semesters, total $34,000.

Hansung Scholarship, Hansung Son Jae Han Foundation 2019.03 - 2021.02
$8,000 over two years.

Alumni Association Award, Daegu Science High School 2021.02
Graduated with the highest honors.

Talks

1. "IDT: Intelligent Data Placement for Multi-tiered Main Memory with Reinforcement Learning", **Oral Presentation**, *HPDC*, Pisa, Italy, 2024
2. "Searching Optimal Compiler Optimization Passes Sequence for Reducing Runtime Memory Profile using Ensemble Reinforcement Learning", Short Talk, *EMSOFT*, Hamburg, Germany, 2023
3. "Accuracy-Area Efficient Online Fault Detection for Robust Neural Network Software-Embedded Microcontrollers", Short Talk, *EMSOFT*, Shanghai, China (Virtual), 2022

Mandatory Military Service

Defense Communication Command, Republic of Korea Army 2024.07 - 2026.01
Operated the military microwave wireless communication system in an isolated post. Served as a squad leader and was awarded as an exemplary soldier of the command.